# List of Lab Experiments for CS-368 Reinforcement Learning

| Lab No. | Lab Experiments | CLO-01 | CLO-02 | CLO-03 | CLO-04 |
|---|---|---|---|---|---|
| Lab 1 | Introduction to NumPy Library | ✓ | | | ✓ |
| Lab 2 | Data Visualization in Python | ✓ | | | ✓ |
| Lab 3 | Markov Decision Processes – Iterative Policy Evaluation, Policy Iteration, Value Iteration | ✓ | | | ✓ |
| Lab 4 | Monte Carlo & TD Learning – Prediction [Cliff Walking & Frozen Lake Env.] | ✓ | | | ✓ |
| Lab 5 | n-Step TD and TD (lambda) – Prediction [Frozen Lake & Taxi Environments] | ✓ | | | ✓ |
| Lab 6 | SARSA – Control [Cart Pole & Acrobat Environments] | ✓ | | | ✓ |
| Lab 7 | n-Step SARSA & SARSA (Lambda) – Control [Cart Pole & Pendulum Environments] | ✓ | | | ✓ |
| Lab 8 | Q-Learning & Double Q-Learning – Control [Mountain Car & Lunar Landar Environments] | ✓ | | | ✓ |
| Lab 9 | Deep Q Networks – I [Acrobat Environment] | ✓ | | | ✓ |
| Lab 10 | Deep Q Networks – II (Stable Baselines3) [Lunar Lander & Blackjack Environments] | ✓ | | | ✓ |
| Lab 11 | Advantage Actor Critic (A3C) [Bipedal Walker & Car Racing Environments] | ✓ | | | ✓ |
| Lab 12 | Proximal Policy Optimization (PPO) [Pendulum & Taxi Environments] | ✓ | | | ✓ |
| Lab 13 | Soft Actor Critic (SAC) [Blackjack Environment] | ✓ | | | ✓ |
| Lab 14 | Open Ended Lab [MuJoCo Environment] | ✓ | | | ✓ |
| Lab 15 | Project [Atari Environments] | | ✓ | ✓ | ✓ |
| Lab 16 | Project [Atari Environments] | | ✓ | ✓ | ✓ |

**Mapping of the Course Learning Outcomes (CLO) to the Program Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:**

| No. | Course Learning Outcomes (verb, condition, standard) | PLO (SE) | PLO (CS/AI/ DS) | BT Level | Teaching & Learning Methods | Assessment Methods |
|---|---|---|---|---|---|---|
| CLO 1 | **Apply** reinforcement learning algorithms (e.g., Dynamic Programming, Temporal Difference Learning, Q-Learning) when given real-world problem scenarios to develop practical solutions. | PLO 1 | PLO 2 | C-3 (Application) | Active learning, Cooperative Learning, Blended Learning | Assignment Project Quiz MSE ESE Labs |
| CLO 2 | **Analyze** reinforcement learning algorithms by evaluating convergence behavior, sample efficiency, and policy performance to determine their relative effectiveness. | PLO 2 | PLO 3 | C-4 (Analysis) | Active learning, Cooperative Learning, Blended Learning | Assignment Project Quiz MSE ESE |
| CLO 3 | **Design** reinforcement learning solutions for complex, real-world inspired tasks by selecting appropriate algorithms and justifying design decisions. | PLO 3 | PLO 4 | C-6 (Create) | Active learning, Cooperative Learning, Blended Learning | Assignment Project MSE ESE Labs |
| CLO 4 | **Implement** reinforcement learning algorithms in simulated and real-world applications using programming languages such as Python and frameworks such as TensorFlow or PyTorch. | PLO 5 | PLO 5 | P-3 (Guided Response) | Cooperative Learning, Blended Learning | Project Labs |

# Lab Rubrics

| Assessment | Does not meet expectation (1/2 marks) | Meets expectation (3/4 marks) | Exceeds expectation (5 marks) |
|---|---|---|---|
| **Software Problem Realization (CLO1, PLO1)** | The student struggles to formulate the problem as RL and does not apply RL prediction algorithms to solve it. There is a lack of understanding of the problem's requirements and no attempt to evaluate the given policy effectively. | The student formulates the problem as RL with some guidance, applies Monte Carlo and TD (0) algorithms with hints, and shows it's working. However, the approach might not be fully optimized or lacks a thorough justification. | The student independently formulates the given problem as RL, applies Monte Carlo and TD (0) algorithms without guidance, and effectively evaluates the given policy. The approach is fully optimized and can be applied to different similar problems. |
| **Software Tool Usage (CLO4, PLO5)** | Code has syntax errors, and the implementation of the Monte Carlo / TD (0) algorithm is incorrect or incomplete. The code is not modular and lacks comments for readability and reuse. The student shows limited ability to use gymnasium library functions where required. | The student has implemented the Monte Carlo and TD (0) algorithms correctly for the given problem with minor mistakes. The code is mostly correct in terms of syntax and functionality but might not be optimized or well-structured for reuse. Some documentation is provided. The student also shows working knowledge of gymnasium library where required. | The student has implemented the Monte Carlo and TD (0) algorithms efficiently and correctly. The code is clean, modular, well-documented, and follows best practices for reproducibility and reuse. The student demonstrates full command of the gymnasium library and its functions. |

## Rubrics Based Evaluation

| Lab 4 | | | | | |
|---|---|---|---|---|---|
| **Reg. No.** | **Name** | **Best/Average/Worst** | **CLO-1** | **CLO-4** | **Total** |
| 406270 | Awais Nazir | Best | 4 | 4 | 8 |
| 407411 | Hafiz Muhammad Ammar | Average | 3 | 4 | 7 |
| 406483 | Muhammad Faiq Qazi | Worst | 3 | 3 | 6 |